

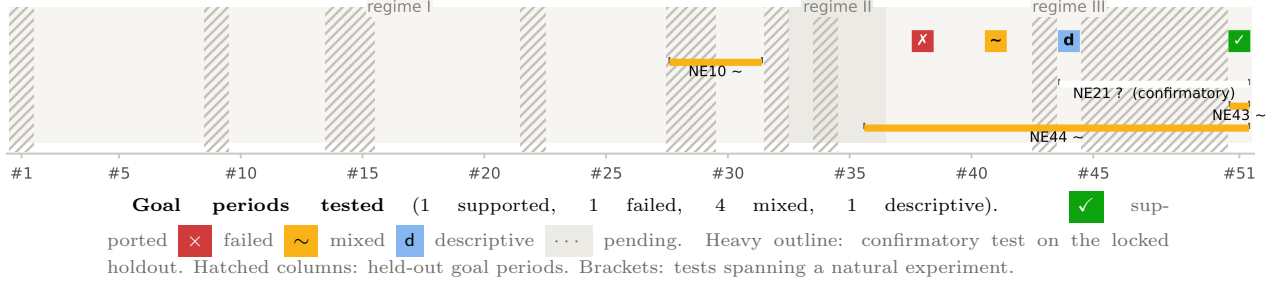
H04: External forcing reshapes the response kernel, reversibly

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 1.5/5 Usefulness 2.5/5

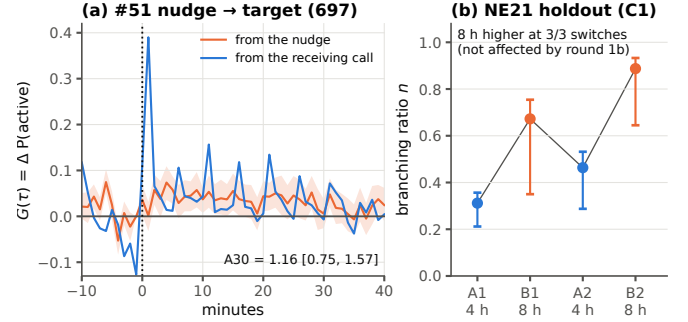
Bottom line. A nudge acts only on its target, at the target's next model call (the apparent minutes-long dead time was a measurement artifact), and no hours or nudger switch lowered the swarm's self-excitation as predicted.

level: hypothesis · pipeline checks 98% · updated 2026-10-04



WHAT HAPPENED

Exploratory round 1b (non-holdout, corrected activity table, leading-@ targets, no future-kick isolation, day fixed effect): a nudge adds 1.16 [0.75, 1.57] active minutes to its named target over 30 min in #51 (first nudges 1.38, repeats 0.90; regime-III 4 h days 0.93), bystanders ≈ 0 . The response starts at the target's next model call (median read-out 108 s; read-out-aligned onset 1 min): round 1's 4-min dead time was dropped events plus the isolation rule. Human messages no longer move activity. FDT fails in shape; kicked activations persist ($X \approx 1.9$); mean-field Glauber underpredicts decay times. Nudger off (NE43): n unchanged, traps not longer. **Locked holdout NE21** (hours 4 $\rightarrow$ 8 $\rightarrow$ 4 $\rightarrow$ 8 h): **C1 falsified in reverse** (n 0.31, 0.67, 0.46, 0.89); C1 does not use the corrected table and stands.



Observables. (a) A nudged agent's activity against matched controls (corrected design, 95% day-block band), aligned on the nudge and on the target's receiving call. Realigned, the response is immediate: the delay seen from the nudge is the wait until the target's next call. (b) Branching ratio per hours segment on the holdout; every switch went the other way, at the edge of week-to-week noise.

PHYSICAL MODEL

Models 02 (linear response, fluctuation-dissipation) and 09 (Hawkes). $n_i(m) \in \{0, 1\}$: agent i active in minute m . A kick (nudge, human message) at m_0 is a field pulse; G : matched response; r : read-out delay to the target's next model call; C : activity autocorrelation; χ : integrated G ; X : FD ratio. Hawkes: $\mu_{ab}(t)$ daily baseline, ϕ talk-triggered kernel, η_c messages triggered per class- c kick, n branching ratio. Mean field: loop gain K , agent time τ_0 .

$$G(\tau) = \mathbb{E}[n_i(m_0 + \tau) \mid \text{kick}] - \mathbb{E}[n_i(m_0 + \tau) \mid \text{control}] - \delta_d(\tau)$$

$$G(\tau) \approx \sum_r p(r) \Delta R(\tau - r), \quad \text{FDT: } \chi(\tau) = X [C(0) - C(\tau)]$$

$$\lambda(t) = \mu_{ab}(t) + \sum_k \phi(t - t_k) + \sum_c \eta_c \sum_j \gamma_c e^{-\gamma_c(t - t_j)}$$

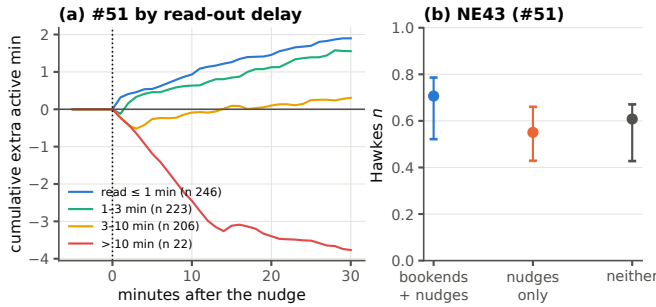
$$n = \int_0^\infty \phi(\tau) d\tau, \quad \tau = \tau_0 / (1 - K)$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The useful part is plain and now sharper. A nudge works on the agent it names and nobody else, at that agent's next model call, so its lag is the target's pause schedule: a nudge to an agent paused for 10 minutes does nothing until it wakes. A first nudge buys about 1.4 active minutes, a repeat less. Human messages change what agents say, not how much they work. The physics vocabulary (Green's functions, FDT, effective temperature) mostly shows that a field-on-activity model is wrong. The one reversal test, longer days lowering self-excitation, failed in the opposite direction; switching the nudger off left self-excitation unchanged.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
P1 nudge: A60 > 0, peak ~15 min → #51 peak 4, yes (#51, 4 h)	A60 1.65 [0.79, 2.52]	yes
Nudge A30 (isolated → 1.54 → 1.16 (#51), 0.93 (4 h); corrected all nudges)	isolation inflates ×1.3–1.9	corrected
First vs repeat nudge (#51)	→ 1.38 vs 0.90	sub-additive
P2 bystanders ≤0.2 of yes → 0.03 (#51) target		yes
P3 human messages mentioned	5.6 → 0.6 (n.s.); room move activity ≈ 0	no
P6 FDT nudge X < 1	violated, violated, X 1.54 → 1.88	violated; dir. no
MF-P1/P3 (K subcritical, self-consistent)	K III 0.26 → 0.32	yes
NE43 nudger off: n Δn +0.06; inactive runs same, traps longer	+0.5%	mixed
NE44 read-out moves with pause default	43 s vs 108 s; onset lag equal from read-out	mixed
C1 (holdout) n lower 0/3 switches; not affected by 1b on 8 h days		falsified



Second figure. (a) Cumulative response of #51 nudge targets by how long the nudge waited to be read: read within 3 min, the response starts at once; read after 10 min (long pauses), the target stays below its controls. (b) NE43: Hawkes n with the daily bookends and nudges, with nudges only, and with neither.

GOAL PERIODS IN DETAIL

NE21 ? 1b: not re-run (locked holdout; re-running is Vivian's call). C1 (H... **(holdout)**)

G38 × failed

G41 ~ mixed

G44 d descriptive

G51 ✓ supported

NE10 ~ mixed

NE43 ~ mixed

NE44 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

The #51 pre-window placebo is +0.25 [0.05, 0.46]: some selection by the nudger's text trigger remains, so the level may be up to a quarter too high. Long pauses are not in the matching strata. Regime-I nudges (NE10) fail the placebo. Hawkes n may absorb within-day nonstationarity. The holdout's C2, C4 and MF-C used the buggy activity table and the isolation rule; re-running them is Vivian's call.

ROUND 2

Add pause state to the matching strata; fit a read-out + transient kernel.

- **Essence.** The scaffold is the propagator: the response to any input is when the agent next looks, times whether it attends.
- **H04-R1** The response kernel equals the turn-interval survival function convolved with an uptake probability, with no free parameters (HH92, tested in H08).
- **H04-R2** The steering knob is the scheduler: shortening turn cadence raises susceptibility more than rewording or repeating a message (E1).
- **H04-R3** A salience law: uptake rises with mention, position and novelty in context and falls with backlog (E5).